

Exact Hive Certificates for the Balanced-Inertia Stability Constants ($N = 3$ to 12) in Every Dimension, the (A_N, B) Cost Law, and the Twenty-Two-Form Six-Level Cost Formula

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14 September 2026 — revised candidate

Abstract

For a traceless Hermitian F with N positive, N negative and $d - 2N$ zero eigenvalues, the inverse self-commutator cost $\kappa_d(F)$ satisfies $\delta_d \geq \gamma_{N,d} D_*$ with an optimal rational constant $\gamma_{N,d}$; the values $\gamma_2, \dots, \gamma_5$ were known and $\gamma_6, \dots, \gamma_{10}$ only numerically. We give exact, independently machine-checked certificates proving $\gamma_{N,d} = \gamma_N$ for $N = 3, \dots, 12$ and every $d \geq 2N$: $17/36, 4/7, 113/152, 9/10, 781/740, 317/264, 23/17, 3/2, 773/468, 289/160$. The minimising pairs switch at $N = 7$ from (A_N, B) to (u_3, z_N) . The certificates are exact rational hive primals at every positive-distance vertex of the finite reduction and matching duals at the listed minimisers, and dimension-independent Horn combinations (Weyl, two size-two and Lidskii–Wielandt inequalities) at the minimisers, written out in full. We also prove the (A_N, B) cost law for $N \leq 14$ and the exact six-level formula $\kappa_6 = \max$ of twenty-two explicit linear forms, all exposed. The supplied package reports a separate adversarial replay; fresh checks using separately written code are documented in Section 7, with provenance limits stated explicitly.

1. Introduction

Antecedents. The quantity studied here is the least Hilbert–Schmidt cost of realising a prescribed traceless Hermitian F as a self-commutator, $\kappa_d(F) = \frac{1}{2} \min\{\|C\|_{HS}^2 : CC^* - C^*C = 2F\}$. Its theory has been developed in a sequence of records of the AI Research Record archive, cited by descriptive keys (identifiers and titles are in the reference list); everything used from them is restated in Section 2. [FourLevel] reduced κ_d to a finite linear program over Horn’s inequalities (its Theorem 2.2, the *Horn program*) and gave the exact four-level formula (its Theorem 3.1: the maximum of four spectral forms). [FiveLevel] gave the twelve-term five-level formula (its Theorem 3.1) together with the *per-chamber certificate template* — a sparse Horn dual for each form as a global lower bound and, on the chamber where that form is the largest, an explicit feasible construction attaining it — which Section 6 follows. [RankOnset] introduced the hive machinery into this line (exact rational hive primals and duals, its Section 6 and Lemma 6.1 on coarse graining) and showed (its Remark 3.3) that zero padding can lower κ_d , which is why “every dimension” is a genuine issue. [Ceiling] proved the sharp inertia ceiling $\kappa_d \leq \frac{N+1}{2}P$ on the balanced stratum (its Theorem 1), the stability inequality $\gamma D_* \leq \delta_d \leq \frac{N-1}{2}D_*$ with the sharp reverse constant (its Theorem 3) and the optimal forward constants $\gamma_2 = 1/2$ and $\gamma_3 = 17/36$ (its Theorems 3 and 4). [Bal44] proved the finite reduction (its Theorem A: $\gamma_{N,d}$ is the minimum of a ratio over an explicit finite rational set W_N , independent of d) and $\gamma_4 = 4/7$ (its Theorem B); its Lemma 3 is the first dimension-independent Horn lower bound at an extremal pair. [Bal55] proved $\gamma_5 = 113/152$ (its Theorem 1) by

enumerating all 191,353 Horn triples of $d = 10$. The published $(m, 2)$ paper [Inertia_m2] listed as its Open question 3 the *numerical* values $\gamma_6 = 9/10, \gamma_7 = 781/740, \gamma_8 = 317/264, \gamma_9 = 23/17, \gamma_{10} = 3/2$, obtained by floating-point hive LPs, “without exact certificates”, and noted the switch of the extremal family at $N = 7$. Externally: Horn’s conjecture, proved by Klyachko [Kl98] and Knutson–Tao [KT99] (saturation), with the hive model expounded by Buch [Bu00] and surveyed by Fulton [Fu00], is the tool; Weyl, Lidskii and Wielandt [We12, Li50, Wi55] supply the classical inequalities that turn out to suffice at the minimisers.

Contribution. (i) Theorem 1: $\gamma_{N,d} = \gamma_N$ exactly for $N = 3, \dots, 12$ and every $d \geq 2N$, with the two minimising pairs and their costs; $N = 3, 4, 5$ reproduce the published constants, $N = 6, \dots, 10$ turn the numerical values of [Inertia_m2] into theorems, and $N = 11, 12$ are new. (ii) Theorem 2: the (A_N, B) cost law $\kappa_{2N}(A_N, -B^{\text{rev}}) = \frac{3}{2} + ([(N-2)^2/4] - 1) \frac{N+2}{4N(N-1)}$ for $N \leq 14$, in every $d \geq 2N$ for $N \leq 6$. (iii) Theorem 3: κ_6 is the maximum of twenty-two explicit rational forms on the ordered trace-zero cone, every form exposed, so that the non-wall facet sequence of the unit ball of κ_d begins 2, 4, 12, 22 for $d = 3, \dots, 6$. (iv) A certificate architecture in which each claim is a finite list of exact rational objects checked by two independent verifiers: hive primals and duals, windowed Horn templates valid in every dimension (with a reflection trick that halves the search), and ray witnesses with global duals for the chambers of κ_6 . (v) The dimension-independent certificates at the minimisers are so short — three to seven inequalities, multipliers 1 or 2 — that they are written out in full in Section 5 and can be checked by hand. The findings of the adversarial review (Section 7), including three reporting errors of the working report, are incorporated.

2. Setting

Throughout $d \geq 2$, Hermitian matrices are $d \times d$, spectra are listed in decreasing order, and x^{rev} is the reversed list. For traceless Hermitian F with spectrum λ , [FourLevel] Theorem 2.2 (also [RankOnset] Proposition 2.2, in the normalisation $s = p/2$ used here) gives the *Horn program*

$$\kappa_d(\lambda) = \min \left\{ \sum_{t=1}^d s_t : s_1 \geq \dots \geq s_d \geq 0, (s, -s^{\text{rev}}, \lambda) \text{ Horn-feasible} \right\}, \quad (2.1)$$

where a triple (α, β, γ) of decreasing real d -vectors is *Horn-feasible* if Hermitian A, B exist with spectra α, β and $\text{spec}(A+B) = \gamma$. (Given C , put $R = CC^*/2, S = C^*C/2$: they have a common spectrum $s \geq 0$, $R-S = F$ and $\frac{1}{2}\|C\|^2 = \sum s_t$; conversely $R, S \geq 0$ with common spectrum and $R-S = F$ come from some C .) Letting s_d be free rather than 0 is harmless: if $(s, -s^{\text{rev}}, \lambda)$ is feasible with $s_d = c$, then $(A - cI, B + cI)$ shows $(s - c\mathbf{1}, \dots)$ is feasible at lower cost, so the minimum has $s_d = 0$.

Horn’s inequalities. For $1 \leq r < d$ let T_r^d be Horn’s recursively defined set of triples (I, J, K) of r -subsets of $\{1, \dots, d\}$: $\sum_I i + \sum_J j = \sum_K k + r(r+1)/2$ and, for every $q < r$ and $(F, G, H) \in T_q^r$, $\sum_{f \in F} i_f + \sum_{g \in G} j_g \leq \sum_{h \in H} k_h + q(q+1)/2$ (elements listed increasingly). The theorem of Klyachko and Knutson–Tao [Kl98, KT99, Fu00] states that (α, β, γ) is Horn-feasible iff $\sum \gamma = \sum \alpha + \sum \beta$ and

$$\sum_{k \in K} \gamma_k \leq \sum_{i \in I} \alpha_i + \sum_{j \in J} \beta_j \quad \text{for all } (I, J, K) \in T_r^d, 1 \leq r < d, \quad (2.2)$$

and that T_r^d is exactly the set of triples whose Littlewood–Richardson coefficient $c_{\lambda(I)\lambda(J)}^{\lambda(K)}$ is positive. The counts are $|T^d| = 3, 12, 41, 142, 522, 2062$ for $d = 2, \dots, 7$ and 8,752 for $d = 8$. In (2.1), $\alpha = s, \beta_j = -s_{d+1-j}$, so the trace condition is $\sum \lambda = 0$ and (2.2) reads $\sum_{k \in K} \lambda_k \leq \sum_{i \in I} s_i - \sum_{j \in J} s_{d+1-j}$.

Hives. An order- d hive is a function h on $\{(i, j) \in \mathbb{Z}^2 : i, j \geq 0, i + j \leq d\}$ such that for every unit rhombus of the triangular lattice $h(o_1) + h(o_2) \geq h(a_1) + h(a_2)$ (o the obtuse, a the acute vertices; there are $3\binom{d}{2}$ rhombi). With $h(0, 0) = 0$ we say h has boundary (α, β, γ) if the increments along $(0, 0) \rightarrow (d, 0)$ are $\alpha_1, \dots, \alpha_d$, along $(d, 0) \rightarrow (0, d)$ are $\beta_1, \dots, \beta_d$ and along $(0, 0) \rightarrow (0, d)$ are $\gamma_1, \dots, \gamma_d$ (the Knutson–Tao/Buch convention; the labelling was calibrated against the explicit Horn LP, Section 7). Knutson–Tao [KT99] prove that for integral data an integral hive exists iff $c_{\alpha\beta}^\gamma > 0$, and by saturation iff the hive polytope is nonempty; since the projection of the hive cone to boundaries and the cone cut out by (2.2) are closed rational polyhedral cones that agree on rational points, they agree on real points:

$$(\alpha, \beta, \gamma) \text{ Horn-feasible} \iff \text{an order-}d \text{ hive with boundary } (\alpha, \beta, \gamma) \text{ exists.} \quad (2.3)$$

Hence $\kappa_d(\lambda)$ is the value of the *hive LP*: minimise $\sum s_i$ over s decreasing ≥ 0 and h a hive with boundary $(s, -s^{\text{rev}}, \lambda)$, with $(d+1)(d+2)/2$ hive variables and d common-spectrum variables (including variables fixed by the boundary constraints), and $3\binom{d}{2}$ rhombus rows instead of $|T^d|$ rows.

The balanced stratum. Fix $N \geq 2$ and $d \geq 2N$, normalise $P = 1$, and write the spectrum of an exact-inertia- $(N, N, d - 2N)$ target as $\lambda = (a, 0^{d-2N}, -b^{\text{rev}})$ with a, b in the ordered simplex $\mathcal{S}_N = \{a_1 \geq \dots \geq a_N \geq 0, \sum a_i = 1\}$ (closure allowed). Following [Ceiling, Bal44]: $e = (1, 0, \dots, 0)$, $u = (1/N, \dots, 1/N)$, $H_N = (N + 1)/2$,

$$E(a) = \|a - e\|_1 = 2(1 - a_1), \quad U(a) = \|a - u\|_1, \quad \delta_d(a, b) = H_N - \kappa_d(a, 0, -b^{\text{rev}}),$$

$$D_*(a, b) = \min\{E(a) + U(b), U(a) + E(b)\},$$

and $\gamma_{N,d}$ is the largest constant with $\gamma_{N,d} D_* \leq \delta_d$ on the stratum. **Theorem A of [Bal44]** (finite reduction): for every fixed $d \geq 2N$,

$$\gamma_{N,d} = \min_{(a,b) \in W_N, D_*(a,b) > 0} \frac{H_N - \kappa_d(a, 0, -b^{\text{rev}})}{D_*(a, b)}, \quad (2.4)$$

where W_N is an explicit finite rational set independent of d (the cut vertices of the cell complex of D_* , constructed by geometry.py of the [Bal44] record; the supplied provenance describes this copy as byte-identical to the record's; the present fresh checks do not authenticate that historical byte comparison). Its sizes are $|W_N| = 22, 89, 267, 628, 1354, 2611, 4615, 7614, 12120, 18385$ for $N = 3, \dots, 12$; exactly two members, (e, u) and (u, e) , have $D_* = 0$. (The working report misprinted 4617 and 7616 for $N = 9, 10$; the certified counts $|W_N| - 2$ were right.)

The two families. For $N \geq 3$ put

$$A_N = \left(\frac{3N-2}{4N}, \left(\frac{N+2}{4N(N-1)}\right)^{N-1}\right), \quad B = \left(\frac{1}{2}, \frac{1}{2}, 0^{N-2}\right), \quad u_3 = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0^{N-3}\right), \quad z_N = \left(\frac{4N-3}{9N}, \frac{4N-3}{9N}, \left(\frac{N+6}{9N(N-2)}\right)^{N-2}\right).$$

Direct computation gives $D_*(A_N, B) = \frac{5N-6}{2N}$ and $D_*(u_3, z_N) = \frac{4(7N-12)}{9N}$, both orientations being equal in each case ($E(u_3) = 4/3$, $U(z_N) = \frac{16N-48}{9N}$, $U(u_3) = \frac{2(N-3)}{N}$, $E(z_N) = \frac{2(5N+3)}{9N}$). For $N \leq 12$ the pairs of Table 1 are members of W_N (cut vertices in the sense of [Bal44] Lemma 2).

3. Statements

All three theorems are computer-assisted: each is reduced, by the arguments of Section 4, to a finite list of exact rational identities and inequalities, which are stored in JSON certificates and were replayed by two

independent verifiers (Section 7). The imported theorems are exactly those of Section 2: the Horn program (2.1), Horn–Klyachko–Knutson–Tao (2.2), the hive equivalence (2.3), and Theorem A (2.4). Nothing else is imported; in particular no floating-point computation enters any proof.

Theorem 1 (balanced stability constants, all dimensions). For $N \in \{3, \dots, 12\}$ and every $d \geq 2N$, $\gamma_{N,d} = \gamma_N$ with the values, minimising pairs and costs of Table 1. For $N \leq 6$ the ratio in (2.4) is attained at (A_N, B) and (B, A_N) , for $N \geq 7$ at (u_3, z_N) and (z_N, u_3) ; at these pairs κ_d has the tabulated value for every $d \geq 2N$.

Table 1: Theorem 1. $\kappa = \kappa_d$ at the minimising pair (every $d \geq 2N$); the last column is the number of pairs of W_N with $D_* > 0$, each carrying an exact hive primal.

N	γ_N	pair	κ	D_*	pairs
3	17/36	(A_3, B)	31/24	3/2	20
4	4/7	(A_4, B)	3/2	7/4	87
5	113/152	(A_5, B)	127/80	19/10	265
6	9/10	(A_6, B)	17/10	2	626
7	781/740	(u_3, z_7)	479/315	148/63	1352
8	317/264	(u_3, z_8)	169/108	22/9	2609
9	23/17	(u_3, z_9)	43/27	68/27	4613
10	3/2	(u_3, z_{10})	49/30	116/45	7612
11	773/468	(u_3, z_{11})	1481/891	260/99	12118
12	289/160	(u_3, z_{12})	101/60	8/3	18383

The switch at $N = 7$. At (A_7, B) the cost is 99/56 and the ratio 125/116; at (A_8, B) , 13/7 and 148/119; both exceed the (u_3, z_N) ratios 781/740, 317/264. The certified minimisers are the pairs at which a hive dual is stored; *uniqueness* of the minimising pair is not certified (another pair of W_N could have the same ratio, with its stored hive primal not tight), and the value of γ_N is unaffected by this.

Theorem 2 ((A_N, B) cost law). For $N = 3, \dots, 14$,

$$\kappa_{2N}(A_N, -B^{\text{rev}}) = \frac{3}{2} + \left(\left\lfloor \frac{(N-2)^2}{4} \right\rfloor - 1 \right) \frac{N+2}{4N(N-1)}, \quad (3.1)$$

i.e. 31/24, 3/2, 127/80, 17/10, 99/56, 13/7, 553/288, 2, 907/440, 47/22, 457/208, 59/26. For $N = 3, 4, 5, 6$ the same value is $\kappa_d(A_N, 0^{d-2N}, -B^{\text{rev}})$ for every $d \geq 2N$. Exact values are also certified at (u_3, z_N) in $d = 2N$ for $N = 6, \dots, 12$ (13/9 for $N = 6$, then the column of Table 1) and at the minimising pairs for $N = 6, \dots, 10$ in $d = 2N+1, 2N+2, 2N+3$ (constant in d). The law (3.1) for general N remains a conjecture; for $N = 4$ it re-derives [Bal44] Lemma 3, and [Bal55] Lemma 3 gives the lower bound with $N-4$ in place of $\lfloor (N-2)^2/4 \rfloor - 1$, tight only for $N \leq 5$.

Theorem 3 (six-level formula). For every traceless Hermitian 6×6 F with decreasing spectrum λ and gaps $g_j = \lambda_j - \lambda_{j+1} \geq 0$ ($j = 1, \dots, 5$),

$$\kappa_6(F) = \max_{r=1, \dots, 22} q_r \cdot g, \quad (3.2)$$

with the twenty-two rational forms q_r of Table 2 (Section 6), a list closed under reversal $g \mapsto g^{\text{rev}}$. Every form is exposed: its dominance chamber $C_r = \{g \geq 0 : (q_r - q_j) \cdot g \geq 0 \ \forall j\}$ is full-dimensional, so q_r is the unique maximiser on a non-empty open set. Consequently the unit ball $\{\kappa_d \leq 1\}$ of the ordered trace-zero cone has exactly 2, 4, 12, 22 non-wall facets for $d = 3, 4, 5, 6$ (the counts for $d = 3, 4, 5$ follow from the closed formulas of [FourLevel] and [FiveLevel]).

4. Certificate architecture and proofs

4.1 Exact hive primals and duals

A *hive primal* for λ in dimension d is a rational decreasing $s \geq 0$ together with a rational hive h (stored as integers h_{int} and a common denominator) with boundary $(s, -s^{\text{rev}}, \lambda)$. Checking it means checking $3\binom{d}{2}$ rhombus inequalities and $3d$ boundary equalities in exact arithmetic; by (2.3) it proves $\kappa_d(\lambda) \leq \sum s_t$. A *hive dual* is a list of nonnegative rational multipliers y on the rhombi, z on the ordering rows $s_k - s_{k+1} \geq 0$, $s_d \geq 0$, and signed multipliers μ on the boundary equalities, such that

$$\sum_t s_t - v = \sum_{\text{rhombi}} y \cdot (\text{rhombus slack}) + \sum z \cdot (\text{ordering slack}) + \sum \mu \cdot (\text{boundary residual}) + \sum_t v_t s_t \quad (4.1)$$

holds as an identity of linear forms in (h, s) with $v \geq 0$; then $\sum s_t \geq v$ on every feasible point, and by (2.3) $\kappa_d(\lambda) \geq v$. When primal cost and dual value coincide, $\kappa_d(\lambda)$ is known exactly. The generators solve the hive LP in floating point (HiGHS dual simplex), rationalise s, h and the multipliers by continued fractions, and re-verify exactly before writing anything; a fallback exact re-solve on the floating-point support exists but was never needed (all failure counters are 0 in every run).

4.2 Dimension-independent lower bounds: windowed Horn templates

A *template* is $(I_0, I_1, J_0, J_1, K_0, K_1, r, \text{refl})$ with $I_0, \dots, K_1 \subseteq \{1, \dots, M\}$, $|I_0| + |I_1| = |J_0| + |J_1| = |K_0| + |K_1| = r$ and $|I_1| + |J_1| = |K_1|$. In dimension $d \geq 2M$ it defines $I = I_0 \cup \{d+1-t : t \in I_1\}$ and likewise J, K : “small” indices $\leq M$ and “large” indices $\geq d+1-M$, which do not overlap. Write $s'_t := s_{d+1-t}$. For $\lambda = (a, 0^{d-2N}, -b^{\text{rev}})$ ($M \geq N$; $a_k := 0$ for $k > N$, $b_t := 0$ for $t > N$) the Horn inequality of the template reads

$$\sum_{k \in K_0} a_k - \sum_{t \in K_1} b_t \leq \sum_{i \in I_0} s_i + \sum_{t \in I_1} s'_t - \sum_{j \in J_0} s'_j - \sum_{t \in J_1} s_t, \quad (4.2)$$

an inequality whose coefficients do not depend on d .

Lemma 4.1 (stability). Fix a template and $d_0 \geq 2M$. Membership $(I, J, K) \in T_r^d$ is a finite conjunction of conditions each of which is affine in d (the sum condition has zero slope; each recursive condition $(F, G, H) \in T_q^r$ contributes an inequality whose d -coefficient is $\#\{f \in F : i_f \text{ large}\} + \#\{g \in G : j_g \text{ large}\} - \#\{h \in H : k_h \text{ large}\}$). If every condition holds at $d = d_0$ and has slope ≤ 0 , the template is a Horn triple in every $d \geq d_0$, and (4.2) holds for every feasible s in every such d .

Proof. For $d \geq 2M$ the sorted order of I is d -independent, so the recursive conditions refer to fixed positions; an affine function of d with slope ≤ 0 that is satisfied at d_0 is satisfied for all $d \geq d_0$. The inequality then holds by the necessity half of (2.2). $\square$

Lemma 4.2 (reflection). (α, β, γ) is Horn-feasible iff $(-\alpha^{\text{rev}}, -\beta^{\text{rev}}, -\gamma^{\text{rev}})$ is (negate A, B) iff (β, α, γ) is (commute the sum). Applied to $(s, -s^{\text{rev}}, \lambda)$ this gives $(s, -s^{\text{rev}}, -\lambda^{\text{rev}})$, and $-\lambda^{\text{rev}} = (b, 0^{d-2N}, -a^{\text{rev}})$. Hence the feasible sets of s for the targets (a, b) and (b, a) coincide, κ_d is the same for both orientations, and the Horn inequalities of the reflected target ($\text{refl} = 1$: exchange the roles of a and b in (4.2)) are valid constraints on the same s .

An *all- d certificate* for (a, b) with bound v is a list of stable templates with nonnegative rational multipliers y (and, optionally, multipliers on the ordering rows, unused in all the certificates here) such that the y -combination of the inequalities (4.2) is $\sum_i c_i s_i + \sum_t c'_t s'_t \geq v$ with every $c_i, c'_t \leq 1$. Since $s \geq 0$, $\sum_{\text{all}} s \geq \sum_i c_i s_i + \sum_t c'_t s'_t \geq v$ in every $d \geq d_0$, i.e. $\kappa_d(a, 0^{d-2N}, -b^{\text{rev}}) \geq v$ for all $d \geq d_0$. The verifier re-derives stability of every template from Horn’s recursion (its own implementation) and re-assembles

the combination; a second checker instantiates each template at $d = d_0, \dots, d_0 + 8$ and tests membership in T_r^d directly.

The certificates were found by linear programming over families of stable templates: a generic enumeration of all stable templates with $r \leq 5$ (`stable_templates.py`; 1.53 million templates for $N = 6$, 496 s) and, much more cheaply, the *structured family* (`all_d_family.py`) consisting of all stable templates with $r \leq 2$ together with all Lidskii–Wielandt templates $I = \{1, \dots, r\}, J = K$ and their mirrors, in both orientations. The structured family reproduces the $N = 6, 7$ certificates in 0.5 s and 1.4 s and certifies $N = 3, 4, 5, 8, \dots, 12$ in at most 74 s each. A generic search restricted to $r \leq 4$ cannot certify $N = 7$ (best bound $466/315 < 479/315$): the tight certificate needs an $r = 6$ Lidskii–Wielandt inequality.

4.3 Chambers of κ_6 : ray witnesses and global duals

Work in $d = 6$ with $s_6 = 0$ (Section 2) and gap coordinates $g \in \mathbb{R}_{\geq 0}^5$, $\lambda(g)$ the trace-zero spectrum with those gaps; (2.1) becomes the explicit LP with the 522 rows of T^6 and four ordering rows. For a form q two objects are stored. (a) A *global dual*: multipliers $y \geq 0$ on the 522 triples and $z \geq 0$ on the ordering rows with $c := \sum_r y_r \alpha_r + D^\top z \leq (1, 1, 1, 1, 1)$ componentwise, where α_r is the s -coefficient vector of row r and D the ordering matrix, and $\sum_r y_r \mathbf{1}_{K_r} = q$ as a linear form on trace-zero λ (i.e. in gap coordinates). Summing the rows gives $c \cdot s \geq q \cdot g$; since $s \geq 0$ and $c \leq \mathbf{1}$ componentwise, $\sum s \geq c \cdot s \geq q \cdot g$ for every feasible s and every g , since the multipliers do not depend on λ ; hence $\kappa_6 \geq \max_r q_r \cdot g$ everywhere. (b) *Ray witnesses*: the extreme rays ρ of the dominance chamber C_r (primitive integer vectors) and, at each, an exact feasible $s(\rho)$ with $\sum s(\rho) = q_r \cdot \rho$. The feasible set $\{(s, \lambda)\}$ of the Horn program is a polyhedral convex cone, $\lambda(g)$ is linear, and $C_r \subseteq \mathbb{R}_{\geq 0}^5$ is pointed, hence generated by its extreme rays; for $g = \sum c_\rho \rho$ with $c \geq 0$ the point $\sum c_\rho s(\rho)$ is feasible for $\lambda(g)$ with cost $q_r \cdot g$. So $\kappa_6 \leq q_r \cdot g$ on C_r , and with (a), $\kappa_6 = q_r \cdot g = \max_j q_j \cdot g$ there. Coverage is automatic: for every g some r attains the maximum and then $g \in C_r$ by definition. Extreme rays were computed by brute force over 4-subsets of the chamber inequalities (complete for a pointed cone in $\mathbb{R}^5$: an extreme ray has at least four linearly independent tight inequalities) and, in review, by `pycddlib` (exact GMP arithmetic); the two sets coincide.

4.4 Padding monotonicity and the proof of Theorem 1

Lemma 4.3 (padding monotonicity). If C realises F in dimension d then $C \oplus 0_{e-d}$ realises $F \oplus 0_{e-d}$ with the same Hilbert–Schmidt norm; the sorted spectrum of $F \oplus 0$ is $(a, 0^{e-2N}, -b^{\text{rev}})$ (the new zeros land between $a_N \geq 0$ and $-b_N \leq 0$). Hence $\kappa_e(a, 0^{e-2N}, -b^{\text{rev}}) \leq \kappa_d(a, 0^{d-2N}, -b^{\text{rev}})$ for $e \geq d \geq 2N$; the ratios in (2.4) are non-decreasing in d and $\gamma_{N,d} \geq \gamma_{N,2N}$. $\square$

Proof of Theorem 1. Fix N and let m be the value in Table 1. (i) By Theorem A, $\gamma_{N,d}$ is the minimum of (2.4) over the $|W_N| - 2$ pairs with $D_* > 0$, for every fixed $d \geq 2N$. (ii) For every such pair p the stored order- $2N$ hive primal gives $\kappa_{2N}(p) \leq \sum s(p)$, and by Lemma 4.3 $\kappa_d(p) \leq \kappa_{2N}(p)$; the verifier checks $(H_N - \sum s(p))/D_*(p) \geq m$ for every p . So every ratio in every $d \geq 2N$ is $\geq m$. (iii) At the two minimising pairs p^* the stored hive dual gives $\kappa_{2N}(p^*) = \sum s(p^*)$ exactly, and the all- d certificate of Section 5 gives $\kappa_d(p^*) \geq \kappa_{2N}(p^*)$ for every $d \geq 2N$; with Lemma 4.3, $\kappa_d(p^*) = \kappa_{2N}(p^*)$ and the ratio at p^* equals m in every d . By Lemma 4.2 one all- d certificate covers both orientations. Hence $\gamma_{N,d} = m$ for all $d \geq 2N$. $\square$

Which imported facts are used where: the upper bounds (ii) use the hive equivalence (2.3) in the direction “hive $\Rightarrow$ feasible” in dimension $2N$ only; the exact values at the minimisers use (2.3) in the direction “feasible $\Rightarrow$ hive” in dimension $2N$; the all- d lower bounds use only the *necessity* of Horn’s inequalities, in every d . The inequalities that occur (Section 5) are Weyl’s ($r = 1$), two $r = 2$ types with Littlewood–Richardson coefficient 1 (by the Pieri rule, e.g. $c_{(2),(d-4)}^{(d-4,2)}$ for $I = \{1, 4\}, J = \{1, d-2\}, K = \{3, d-2\}$), their mirrors, and Lidskii–Wielandt inequalities ($I = \{1, \dots, r\}, J = K$), all classical [We12, Li50, Wi55].

Proof of Theorem 2. Each of the 34 entries of `AB_costs.json` carries a hive primal and a hive dual of equal value (Section 4.1), which is (3.1) for the twelve (A_N, B) entries. For $N \leq 6$ the all- d certificates `alld_lower_AB_N3..N6` give $\kappa_d \geq$ the value in every $d \geq 2N$, and Lemma 4.3 gives $\leq$. $\square$

Proof of Theorem 3. Section 4.3 with the stored 22 duals and 112 ray witnesses; exposedness is the rank-5 check on each chamber. $\square$

5. The explicit all- d certificates

Notation: $s'_t = s_{d+1-t}$; each line is one Horn inequality (4.2), valid for every feasible s in every $d \geq 2N$ by Lemmas 4.1–4.2; “refl” means the inequality is taken for the reflected target $(b, 0, -a^{\text{rev}})$; multipliers are written in front. The index sets below are those of the JSON files (the working report misprinted the J and K sets of the $r = 2$ templates for $N = 5, 9, 12$; the inequalities themselves were right). All lines are reproduced, with their exact values, by `print_alld_certificates.py`.

$N = 6, (A_6, B), a = (2/3, (1/15)^5), b = (1/2, 1/2, 0^4), d \geq 12$:

- refl Weyl, $I = K = \{2\}, J = \{1\}$: $b_2 = \frac{1}{2} \leq s_2 - s'_1$;
- Weyl, $I = K = \{6\}, J = \{1\}$: $a_6 = \frac{1}{15} \leq s_6 - s'_1$;
- refl Lidskii–Wielandt $r = 3, I = \{1, 2, 3\}, J = K = \{1, 2, d-1\}$: $b_1 + b_2 - a_2 = \frac{14}{15} \leq (s_1 + s_2 + s_3) - (s'_1 + s'_2 + s_2) = s_1 + s_3 - s'_1 - s'_2$;
- refl Lidskii–Wielandt $r = 5, I = \{1, \dots, 5\}, J = K = \{1, 2, d-2, d-1, d\}$: $b_1 + b_2 - a_1 - a_2 - a_3 = \frac{1}{5} \leq \sum_{i \leq 5} s_i - (s'_1 + s'_2 + s_1 + s_2 + s_3) = s_4 + s_5 - s'_1 - s'_2$.

Sum: $s_1 + \dots + s_6 - 4s'_1 - 2s'_2 \geq \frac{1}{2} + \frac{1}{15} + \frac{14}{15} + \frac{1}{5} = \frac{17}{10}$; all coefficients ≤ 1 and $s \geq 0$ give $\sum_{\text{all}} s \geq 17/10 = \kappa_{12}(A_6, B)$, so $\kappa_d = 17/10$ for all $d \geq 12$.

$N = 7, (u_3, z_7), a = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0^4), b = (\frac{25}{63}, \frac{25}{63}, (\frac{13}{315})^5), d \geq 14$:

- $2 \times$ Weyl, $I = K = \{3\}, J = \{1\}$: $2a_3 = \frac{2}{3} \leq 2s_3 - 2s'_1$;
- refl Weyl, $I = K = \{7\}, J = \{1\}$: $b_7 = \frac{13}{315} \leq s_7 - s'_1$;
- $r = 2, I = \{1, 4\}, J = \{1, d-2\}, K = \{3, d-2\}$: $a_3 - b_3 = \frac{92}{315} \leq s_1 + s_4 - s'_1 - s_3$;
- Lidskii–Wielandt $r = 6, I = \{1, \dots, 6\}, J = K = \{1, 2, 3, d-3, d-2, d\}$: $a_1 + a_2 + a_3 - b_1 - b_3 - b_4 = \frac{164}{315} \leq \sum_{i \leq 6} s_i - (s'_1 + s'_2 + s'_3 + s_1 + s_3 + s_4) = s_2 + s_5 + s_6 - s'_1 - s'_2 - s'_3$.

Sum: $s_1 + \dots + s_7 - 5s'_1 - s'_2 - s'_3 \geq \frac{210+13+92+164}{315} = \frac{479}{315} = \kappa_{14}(u_3, z_7)$.

The remaining certificates (same conventions; W_k = Weyl on index k , i.e. $\lambda_k \leq s_k - s'_1$; L_6 = the $r = 6$ Lidskii–Wielandt line above; L_9 = Lidskii–Wielandt $r = 9$ with $I = \{1, \dots, 9\}, J = K = \{1, 2, 3\} \cup \{d-5, \dots, d\}$: $a_1 + a_2 + a_3 - \sum_{t \leq 6} b_t \leq s_7 + s_8 + s_9 - s'_1 - s'_2 - s'_3$; R_2 = the $r = 2$ line $I = \{1, 4\}, J = \{1, d-2\}, K = \{3, d-2\}$ above):

- $N = 3, (A_3, B)$, bound $31/24$: $W_1 (a_1 = \frac{7}{12}), \text{refl } W_2 (b_2 = \frac{1}{2}), W_3 (a_3 = \frac{5}{24})$; sum $s_1 + s_2 + s_3 - 3s'_1 \geq 31/24$.
- $N = 4, (A_4, B)$, bound $3/2$: $2 \times \text{refl } W_2, W_4 (a_4 = \frac{1}{8}), \text{refl } r = 2$ with $I = \{1, 3\}, J = \{1, d-1\}, K = \{2, d-1\}$: $b_2 - a_2 = \frac{3}{8} \leq s_1 + s_3 - s'_1 - s_2$ (the two-row argument of [Bal44] Lemma 3); sum $s_1 + \dots + s_4 - 4s'_1 \geq 3/2$.

- $N = 5, (A_5, B)$, bound $127/80$: $2 \times \text{refl } W_2, W_5$ ($a_5 = \frac{7}{80}$), the $\text{refl } r = 2$ line of $N = 4$ ($b_2 - a_2 = \frac{33}{80}$), and $r = 2$ with $I = \{4, d - 4\}, J = \{1, 5\}, K = \{5, d - 2\}$: $a_5 - b_3 = \frac{7}{80} \leq s_4 + s'_5 - s'_1 - s'_5 = s_4 - s'_1$; sum $s_1 + \dots + s_5 - 5s'_1 \geq 127/80$.
- $N = 8, (u_3, z_8)$, bound $169/108$: the $N = 7$ certificate with W_7 replaced by $\text{refl } W_7$ and $\text{refl } W_8$ ($b_7 = b_8 = \frac{7}{216}$), R_2 ($\frac{65}{216}$), L_6 ($\frac{115}{216}$); sum $s_1 + \dots + s_8 - 6s'_1 - s'_2 - s'_3$.
- $N = 9, (u_3, z_9)$, bound $43/27$: $2 \times W_3$, $\text{refl } W_8$, $\text{refl } W_9$ ($\frac{5}{189}$ each), R_2 ($\frac{58}{189}$), $\text{refl } r = 2$ with $I = \{7, d - 8\}, J = \{1, 9\}, K = \{9, d - 3\}$: $b_9 - a_4 = \frac{5}{189} \leq s_7 + s'_9 - s'_1 - s'_9 = s_7 - s'_1$ (weaker than Weyl, chosen by the LP for the coefficient budget), L_6 ($\frac{34}{63}$); sum $s_1 + \dots + s_9 - 7s'_1 - s'_2 - s'_3$.
- $N = 10, (u_3, z_{10})$, bound $49/30$: $2 \times W_3$, $\text{refl } W_{10}$ ($\frac{1}{45}$), R_2 ($\frac{14}{45}$), L_6 ($\frac{49}{90}$), L_9 ($\frac{4}{45}$); sum $s_1 + \dots + s_{10} - 6s'_1 - 2s'_2 - 2s'_3$.
- $N = 11, (u_3, z_{11})$, bound $1481/891$: $2 \times W_3$, $\text{refl } W_{10}$, $\text{refl } W_{11}$ ($\frac{17}{891}$ each), R_2 ($\frac{280}{891}$), L_6 ($\frac{488}{891}$), L_9 ($\frac{85}{891}$); sum $s_1 + \dots + s_{11} - 7s'_1 - 2s'_2 - 2s'_3$.
- $N = 12, (u_3, z_{12})$, bound $101/60$: $2 \times W_3$, $\text{refl } W_{11}$, $\text{refl } W_{12}$ ($\frac{1}{60}$ each), R_2 ($\frac{19}{60}$), $\text{refl } r = 2$ with $I = \{10, d - 11\}, J = \{1, 12\}, K = \{12, d - 3\}$: $b_{12} - a_4 = \frac{1}{60} \leq s_{10} - s'_1$, L_6 ($\frac{11}{20}$), L_9 ($\frac{1}{10}$); sum $s_1 + \dots + s_{12} - 8s'_1 - 2s'_2 - 2s'_3$.

In every case the multipliers are 1 or 2, no ordering multiplier is needed, and the assembled left side is $\sum_{i \leq N} s_i$ minus nonnegative multiples of s'_1, s'_2, s'_3 . The $N = 7, \dots, 12$ certificates share a skeleton — $2 \times$ Weyl on λ_3 , reflected Weyl on the flat tail of z_N , the $r = 2$ line R_2 , and Lidskii-Wielandt lines with $K = \{1, 2, 3\} \cup$ large indices ($r = 6$ for $N \leq 9$; $r = 6$ and 9 for $N \geq 10$).

6. The κ_6 formula

Table 2 lists the twenty-two forms in gap coordinates, the index of the reversed form, the number of extreme rays of the dominance chamber and the number of Horn triples in the support of the global dual (no ordering multipliers are needed in any of the 22 duals). The list is the eleven representatives found by LP-dual facet discovery in the author's cycle-1 notes together with their reversals; the certificate generator, which would have appended a missing facet whenever a ray witness failed, never had to. There are 112 rays in total.

Table 2: the forms q_r of Theorem 3 ($\kappa_6 = \max_r q_r \cdot g$, g the five gaps).

r	q_r	rev	rays	dual
1	$(-3/2, -2, -3/2, 0, 5/2)$	22	5	5
2	$(-7/6, -4/3, -1/2, 4/3, 13/6)$	21	5	4
3	$(-5/6, -2/3, 1/2, 5/3, 11/6)$	20	5	4
4	$(-1/2, 0, 1/2, 2, 3/2)$	19	6	3
5	$(-1/2, 0, 3/2, 1, 3/2)$	18	5	3
6	$(-1/2, 0, 3/2, 2, 1/2)$	13	5	3
7	$(-1/6, 2/3, 3/2, 4/3, 7/6)$	17	5	3
8	$(0, 0, 1, 2, 1)$	16	5	4
9	$(1/6, 1/3, 3/2, 5/3, 5/6)$	15	5	3
10	$(1/6, 4/3, 3/2, 2/3, 5/6)$	14	5	3
11	$(1/2, 0, 1/2, 2, 1/2)$	12	5	3
12	$(1/2, 2, 1/2, 0, 1/2)$	11	5	3

r	q_r	rev	rays	dual
13	$(1/2, 2, 3/2, 0, -1/2)$	6	5	3
14	$(5/6, 2/3, 3/2, 4/3, 1/6)$	10	5	3
15	$(5/6, 5/3, 3/2, 1/3, 1/6)$	9	5	3
16	$(1, 2, 1, 0, 0)$	8	5	4
17	$(7/6, 4/3, 3/2, 2/3, -1/6)$	7	5	3
18	$(3/2, 1, 3/2, 0, -1/2)$	5	5	3
19	$(3/2, 2, 1/2, 0, -1/2)$	4	6	3
20	$(11/6, 5/3, 1/2, -2/3, -5/6)$	3	5	4
21	$(13/6, 4/3, -1/2, -4/3, -7/6)$	2	5	4
22	$(5/2, 0, -3/2, -2, -3/2)$	1	5	5

In spectral coordinates $q \cdot g = \sum_j (q_j - q_{j-1})\lambda_j$ (with $q_0 = q_6 = 0$); e.g. $q_{16} \cdot g = \lambda_1 + \lambda_2 - \lambda_3 - \lambda_4$ and $q_{22} \cdot g = \frac{5}{2}\lambda_1 - \frac{5}{2}\lambda_2 - \frac{3}{2}\lambda_3 - \frac{1}{2}\lambda_4 + \frac{1}{2}\lambda_5 + \frac{3}{2}\lambda_6$. Two exact consistency checks tie the formula to the rest of the paper: at the 20 pairs of W_3 the maximum of the 22 forms equals the hive-certified κ_6 of `gamma_N3.json`, and on 40 random rational spectra it equals the exact primal-equals-dual hive value (`crosscheck_horn_vs_hive.py`). The observation of [FiveLevel] that the facet normals have denominators dividing d persists at $d = 6$ (denominators 1, 2, 3, 6).

7. Verification

Author’s verifiers and supplied historical runs. `verify_certificates.py` (Python standard library, fractions only, no LP solver) replays every certificate exactly: it regenerates W_N from `geometry.py`, recomputes D_* , checks that the certified pairs are exactly those with $D_* > 0$, checks every hive primal with its own rhombus enumeration, every hive dual as the identity (4.1) (in the author’s “stationarity residual” formulation), the minimum ratio and the minimiser list, all 34 `AB_costs` entries, the 522 Horn triples of T^6 from Horn’s recursion, the 22 chambers (rays by its own 4-subset brute force, rank, witnesses, duals), and all 12 active all- d files (stability of each template by its own recursion, assembly, coefficients ≤ 1). Final run: TOTAL: 0 failures, 451 s (`logs/verify_final.log`). `check_templates_explicit.py` instantiates every template at $d = d_0, \dots, d_0 + 8$ (up to $d = 32$ for $N = 12$), tests membership in T_r^d directly from the recursive definition and re-assembles the inequality at each d ; it also tests the sign convention of (2.2) on 130 random Hermitian triples (522 and 2062 inequalities): ALL OK, 18 s. `crosscheck_horn_vs_hive.py` ties the hive LP to the explicit Horn LP of $d = 6, 7, 8$ (522, 2062, 8,752 rows, HiGHS): 0 mismatches, $\max |\text{diff}| \leq 4.4 \cdot 10^{-15}$, 6 s. The generator `hive_exact.py` was itself calibrated against the Horn LP ($\max |\text{diff}| = 2 \cdot 10^{-15}$ on $d = 4, 5$), which fixes the boundary labelling of Section 2.

Author-supplied historical review (described by the author as a separate agent, 2026-09-13; scripts in `repro/reviewer/`). The session identity, model configuration and contextual independence of this historical review have not been authenticated in the present cycle. The following recounts the supplied report and logs. The reviewer wrote its own tools *before* running the author’s: a recursive Horn-triple generator; a Littlewood–Richardson coefficient counter (tableau rule) as a second, non-recursive membership test; an explicit Horn LP for $d \leq 7$; a hive LP in a different formulation (h -variables only, s eliminated); an exact hive-primal checker with rhombi derived from the three lattice directions; an exact hive-dual checker verifying (4.1) as an identity of linear forms (opposite sign convention for μ ; perturbing one multiplier makes it reject); an all- d template checker re-deriving Lemma 4.1 and instantiating at $d = 2N, 2N + 1, 2N + 5, 2N + 9$ with membership tested by recursion and by LR positivity;

and a chamber checker with extreme rays recomputed by pycddlib and by its own brute force. Findings: T_r^d agrees with LR positivity over all sum-condition triples for $d = 5, 6, 7$ (196, 897, 4366 triples, 0 mismatches); the reviewer's hive LP agrees with its Horn LP to $2.7 \cdot 10^{-15}$; every hive primal at all 20, ..., 18383 pairs and every minimiser dual pass its checker, and the minimum ratios and minimiser lists are reproduced; all 34 AB_costs entries have primal = dual; all 12 active all- d files pass; the 112 rays coincide between cdd and brute force, every witness and every dual is valid; on 5,000 random spectra $|\text{Horn LP} - \max_r q_r \cdot g| \leq 1.1 \cdot 10^{-14}$; the author's verifier re-run by the reviewer gave 0 failures in 438.5 s. The reviewer's verdict on all three theorems was CORRECT, with three reporting errors in the working report, corrected here: the sizes $|W_9| = 4615$, $|W_{10}| = 7614$; the J, K sets of the $r = 2$ templates for $N = 5, 9, 12$ (Section 5); and the remark that the gamma_N*.json files store a kappa field for every pair while a dual is stored only at the minimisers — for a non-minimising pair only kappa_upper (the primal cost) is certified, and kappa there is an uncertified record of the generator's dual value.

Supplied rerun of 2026-09-13. repro/run_fast.py (Section 9) re-executed the author's verifier on the subset $N \leq 8$ plus AB_costs, kappa6 and all all- d files, the explicit- d check, the crosscheck, the two manuscript scripts and the reviewer's scripts rev_A, rev_C, rev_E, rev_F and rev_B (default range $N \leq 8$ and AB_costs), all with exit status 0; wall times are in repro/runtimes.txt. The reviewer's logs for $N = 9, \dots, 12$ (rev_B_9_10_11.log, rev_B_12.log) are included unchanged.

Fresh checks of 14 September 2026. The coordinator replayed the supplied full verifier in a separate work copy: all certificates passed, with TOTAL: 0 failures, in 268.969 seconds. A fresh As-tra High task independently reconstructed the finite pair sets, Horn membership using Littlewood-Richardson tableaux, and chamber rays using integer cofactors, without importing the submitted Python code: 47,685 vertex primals, 20 minimiser duals, 34 additional primal-dual entries, 12 active all- d files, 522 dimension-six Horn triples and all 112 chamber-ray incidences passed. A fresh Sol Medium task separately checked the dimension-six chamber certificates and all- d inequalities. Both reviews concerned the original PDF; the underlying certificate bytes and verifier code are unchanged in this corrected candidate. Section 4.3 now states the componentwise coefficient bound actually satisfied by all stored duals (equality is unnecessary). These checks are computational evidence, not human peer review or a proof of model-session independence. Historical scripts and logs remain preserved as historical evidence.

8. Observations and open questions

Observation 8.1 (quasi-polynomial cost at (u_3, z_N) ; conjecture only). Exact primal-equals-dual hive values (not stored, uz_kappa_scan.py, 0.4 s) give, for $N = 6, \dots, 16$, $\kappa_{2N}(u_3, -z_N^{\text{rev}}) = 13/9, 479/315, 169/108, 43/27, 49/30, 1481/891, 101/60, 2206/1287, 73/42, 1028/585, 899/504$. Multiplied by $9N(N-2)$ these are the integers 312, 479, 676, 903, 1176, 1481, 1818, 2206, 2628, 3084, 3596, whose second differences 30, 30, 46 | 32, 32, 51 | 34, 34, 56 are not periodic, but show a pattern across three residue classes compatible with a conjectural quasi-polynomial of period 3 (the three-fold u_3), not a single quadratic. $(15N^2 - 28N - 60)/(9N(N-2))$ fits $N = 6, \dots, 9$ (it is the value of the L_6 certificate pattern of Section 5) and $(16N^2 - 31N - 114)/(9N(N-2))$ fits $N = 9, \dots, 12$; neither fits $N \geq 13$. No closed form is claimed.

Observation 8.2 (conjectured minimiser). For all $N \geq 7$ the minimiser of (2.4) is (u_3, z_N) and $\gamma_N = 9N(H_N - \kappa_{2N}(u_3, z_N))/(4(7N - 12))$; exact for $N \leq 12$, and the ratios at $N = 13, \dots, 16$ would be 6803/3476, 363/172, 913/403, 677/280 if so. The extremal family switches once, at $N = 7$, from the (A_N, B) pairs of [Ceiling, Bal44, Bal55]; whether it switches again (to $(u_4, z_{4,N})$ -type cut vertices) is open.

Open questions. (1) γ_N for all N : a proof that the argmin over W_N lies in the $(u_k, z_{k,N})$ family, and a closed form for $\kappa_{2N}(u_3, z_N)$; the uniform certificate skeleton of Section 5 suggests that the Lidskii-

Wielandt index sets can be given as a function of N (which r occur, and where the large indices sit), but this was not attempted. (2) The general- N cost law (3.1) and its all- d version for $N \geq 7$. (3) The non-wall facets of the unit ball of κ_7 (the sequence 2, 4, 12, 22, ...): the per-chamber template scales, but the 2062-row LP and the number of chambers grow. (4) A representation-theoretic derivation of the 22 facets: the unit ball is the Kirwan polytope of the $U(6)$ -action on $\mathbb{P}(\mathfrak{gl}_6)$, so its facets should be computable by Ressayre-type or Berenstein–Sjamaar restriction criteria rather than by projecting the Horn program.

9. Reproducibility

The directory `repro/` next to this file contains the certificates (`certs/`, 7.5 MB: `gamma_N3.json`, `gamma_N4.json` plain, `gamma_N5..N12.json.gz` from 12 kB to 3.3 MB — uncompressed up to 37 MB —, `AB_costs.json` 230 kB, `kappa6_certificate.json` 11 kB, twelve active `alld_lower_*.json` under 1.3 kB each), the author’s generators and verifiers copied unchanged from the working directory (`crosscheck_horn_vs_hive.py` with one import path changed to find `check_horn_lp.py`, copied from the cycle-1 scratch directory), the author’s logs (`logs/`), the reviewer’s scripts `reviewer/rev_*.py` (unchanged except that two absolute paths were made relative) with the reviewer’s logs, three scripts written for this manuscript (`print_alld_certificates.py`, `kappa6_table.py`, `run_fast.py`) and `README.md` with file sizes, commands, expected final lines and measured wall times. Dependencies: Python 3.12; `numpy` and `scipy` (HiGHS) for the generators, the crosscheck and the reviewer’s LPs; `pycddlib` for `rev_E.py`; the verifiers need only the standard library.

Commands (from `repro/`): `python verify_certificates.py` (full replay, about 7.5 min); `python check_templates_explicit.py` (18 s); `python crosscheck_horn_vs_hive.py` (8 s); `python run_fast.py` (the subset of Section 7, writes `<name>.out` and `runtimes.txt`); `python reviewer/rev_B.py 9 10 11 12 (20, 44, 88, 139 s)`; `python gen_gamma.py N` regenerates a certificate (12 s for $N = 7$ up to 607 s for $N = 12$); `python gen_AB_costs.py` (0.7 s); `python gen_kappa6.py` (106 s); `python alld_family.py N N 2 UZ --lean` regenerates an all- d certificate (≤ 74 s; AB for $N \leq 6$). Expected final lines: TOTAL: 0 failures; ALL OK: True; problems: NONE for every reviewer file; ALL OK from the manuscript scripts. Generating everything took about 40 minutes on one machine, dominated by $N = 10, \dots, 12$ and by the two generic template searches, which the structured family makes unnecessary.

AI-assistance statement

The author supplied the original manuscript, code and certificates as work produced with Claude Fable 5.1, with authoring, adversarial-review and writing roles described in the delivery. These are author-supplied historical identities and role descriptions; neither the serving configuration nor isolation of those sessions has been authenticated here. On 14 September 2026, native Codex reviews configured as GPT-6 Astra High and GPT-5.6 Sol Medium assessed the original exact PDF in fresh delegated tasks and ran separate exact checks. The coordinator then prepared this corrected candidate, including the componentwise dual condition in Section 4.3 and reporting corrections. Those original-PDF assessments are preserved separately and do not constitute assessments or acceptance of this new PDF. No independent human peer review is claimed. The mathematical content rests on the exact certificates and the imported results listed in Section 3. Scientific responsibility rests with the author.

References

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- [FiveLevel] L. Eriksson, *The Exact Five-Level Inverse Commutator Cost: Twelve Horn Chambers, Optimal Rank, and the Sharp 5/2 Resource Tax*, AI Research Record, ARR-2026-37B8R0QTA894GTFF (2026). Cited: Theorem 3.1 (five-level formula) and its per-chamber certificate template.
- [RankOnset] L. Eriksson, *Sharp Onset and Unbounded Growth of Norm-Optimal Self-Commutator Rank*, AI Research Record, ARR-2026-5QQF95VHTC9GABH8 (2026). Cited: Proposition 2.2 (Horn program in the s -normalisation), Remark 3.3 (padding example), Section 6 and Lemma 6.1 (hive certificates and coarse graining).
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